ROLE OF SOLAR ENERGY HARVESTING IN HYBRID ENERGY SOLUTIONS FOR WEARABLE SPORTS ELECTRONICS

Lappeenranta–Lahti University of Technology LUT

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ABSTRACT

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ROLE OF SOLAR ENERGY HARVESTING IN HYBRID ENERGY SOLUTIONS FOR WEARABLE SPORTS ELECTRONICS

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This thesis examines the role of solar energy harvesting within hybrid energy systems for wearable sports electronics. The growing functional demands of modern sports wearables create an increasing mismatch with the limitations of conventional battery-based power supply, motivating the search for supplementary energy sources.

The study reviews solar harvesting fundamentals, hybrid system architectures, and the key factors that govern photovoltaic output at miniaturized scales. These include irradiance variability, device orientation, and available surface area. A scenario analysis of three commercial Garmin solar-assisted devices is conducted to evaluate real-world harvesting performance across representative use conditions. The results confirm that solar harvesting functions primarily as a battery lifetime extender rather than an autonomous power source. Its practical contribution varies substantially with irradiance conditions, device design philosophy and usage intensity. Seasonal asymmetry at northern latitudes further limits the reliability of solar gains outside summer months. Future development in flexible photovoltaic materials and adaptive energy management is expected to improve the viability of solar-assisted wearable systems.

TIIVISTELMÄ

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AURINKOENERGIAN ROOLI HYBRIDI-ENERGIARATKAISUISSA PUETTAVASSA URHEILUELEKTRONIIKASSA

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Tämä kandidaatintyö tarkastelee aurinkoenergian keruun roolia urheilukäyttöön tarkoitettujen puettavien elektroniikkalaitteiden hybridienergiajärjestelmissä. Nykyaikaisten urheilukellojen ja muiden puettavien laitteiden kasvavat toiminnalliset vaatimukset suurentavat kuilua energiantarpeen ja perinteisten akkupohjaisten virtaratkaisujen rajoitteiden välillä. Tämä on väistämättä luonut tarpeen tutkia täydentäviä energianlähteitä.

Työssä käsitellään aurinkoenergian keruun perusteita, hybridijärjestelmien arkkitehtuuria, sekä pienikokoisten aurinkokennojen sähköntuotantoon vaikuttavia keskeisiä tekijöitä. Näitä ovat esimerkiksi säteilyolosuhteiden vaihtelu, laitteen asento, sekä käytettävissä oleva pinta-ala. Lisäksi työssä toteutetaan skenaarioanalyysi kolmelle kaupalliselle Garminin aurinkolatausta hyödyntävälle laitteille, jonka avulla arvioidaan aurinkoenergian keruun todellista suorituskykyä erilaisissa käyttöolosuhteissa. Tulokset osoittavat, että aurinkoenergian keruu toimii ensisijaisesti akun käyttöaikaa pidentävänä ratkaisuna, ei itsenäisenä virtalähteenä. Sen käytännön hyötykäytettävyys vaihtelee huomattavasti säteilyolosuhteiden, laitteen suunnittelun ja käyttömäärän mukaan. Pohjoisemmilla leveysasteilla vuodenaikojen välinen vaihtelu luo lisärajoitteen aurinkoenergian hyödyntämiseen ja täten rajoittaa luotettavimman lataushyödyn kesäkuukausille. Tulevaisuudessa joustavien aurinkokennomateriaalien ja mukautuvan energianhallinnan kehityksen voidaan odottaa parantavan pienikokoisten aurinkoavusteisten järjestelmien käyttökelpoisuutta.

SYMBOLS AND ABBREVIATIONS

Roman characters

P	Power	W, mW, μW
E	Energy	J, Wh
η	Efficiency	
G	Irradiance	$W/m^2, mW/cm^2$
T	Temperature °C	
A	Area	cm^2, m^2

Abbreviations

DC	Direct Current
GPS	Global Positioning System
IoT	Internet of Things
MIP	Memory In Pixel (display)
MPPT	Maximum Power Point Tracking
OLED	Organic Light Emitting Diode
PTC	Photovoltaic USA Test condition
PV	Photovoltaic
STC	Standard Test Conditions
Wi-Fi	Wireless Fidelity

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DECLARATIONS

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1 Introduction

Wearable sports electronics have become a fundamental part of modern athletic practice, enabling continuous monitoring of physiological parameters, location and movement during individual's training or competition. As the functional capabilities of these devices expand rapidly, so does their energy demand. This creates growing tension between what users expect from their devices and what conventional battery-based power systems can sustainably deliver. Limited battery capacity, the physical constraints of the compact and lightweight form factors, and the practical inconvenience of frequent recharging represent challenges that motivate the search for new solutions.

Energy harvesting from ambient sources has emerged as a promising strategy to supplement conventional battery power in wearable electronics. Among the available harvesting methods, including mechanical and thermoelectric approaches, solar energy harvesting stands out as particularly relevant for outdoor applications. Photovoltaic conversion offers comparatively high power density under favourable irradiance conditions and does not depend on user motion or body heat. This makes it especially well-suited for sports devices used in outdoor environments.

However, the integration of photovoltaic harvesting into wearable sports electronics introduces its own inevitable set of challenges. The available surface area of wrist-worn or body-worn devices is somewhat limited. Also, solar irradiance varies significantly with time of day, season, weather and device orientation. It is also relevant to note how the physical demands of sports use imposes strict requirements on material durability and biocompatibility. As a result, solar harvesting in this context rarely becomes a standalone solution. It is rather seen as a valuable component within a broader hybrid energy architecture that combines harvesting with energy storage and intelligent power management.

This thesis examines the role of solar energy harvesting within such hybrid energy systems for wearable sports electronics. The work first provides background on the characteristics and energy demand of modern sports wearables, followed by a review of solar harvesting fundamentals including irradiance variability and achievable power output at miniaturized scales. Hybrid system architectures are then analyzed in terms of their components, energy flow strategies and reported performance. Finally, the last chapter applies the established

framework through a scenario analysis of three commercial Garmin solar-assisted devices, assessing how solar harvesting performs across realistic circumstances and what the results imply for device design and consumer expectations at northern latitudes.

2 Background

This Chapter provides the foundation for the thesis. It begins with an overview of wearable sports electronics as a technology category, followed by an examination of the energy demand and power constraints of modern devices. Finally, the limitations of conventional battery-based power supply are discussed, establishing the motivation for hybrid energy system approaches.

2.1 Overview of Sports Electronics

Sports electronics are a set of technology designed to support physical activity, athletic performance, and sport-related monitoring. Wearable technology refers to devices that can be worn on the human body, performing functions such as sensing, data acquisition, connectivity, and interaction with cloud-based services. (Zhang 2024.)

Wearable technology in sports can also be seen as a part of an Internet of Things (IoT) system, where the device operates through a layered structure. This consists of sensors, processing, and network components. The sensor collects data from the athlete using various measurement methods and transmits signals to the processing layer via technical interfaces. The collected information may include physiological, location, motion and environmental condition data, which enables comprehensive monitoring of both exercise conditions and athlete status. In addition, the network layer supports wireless transmission (e.g. via Bluetooth or Wi-Fi) to external devices and cloud service. This allows more advanced data processing beyond the wearable's often quite limited own computing capacity. The layered architecture of wearable systems is illustrated simply in Figure 1. The Figure demonstrates how the sensing, processing and wireless communication components interact with the whole wearable IoT structure. (Seçkin et al. 2023.)

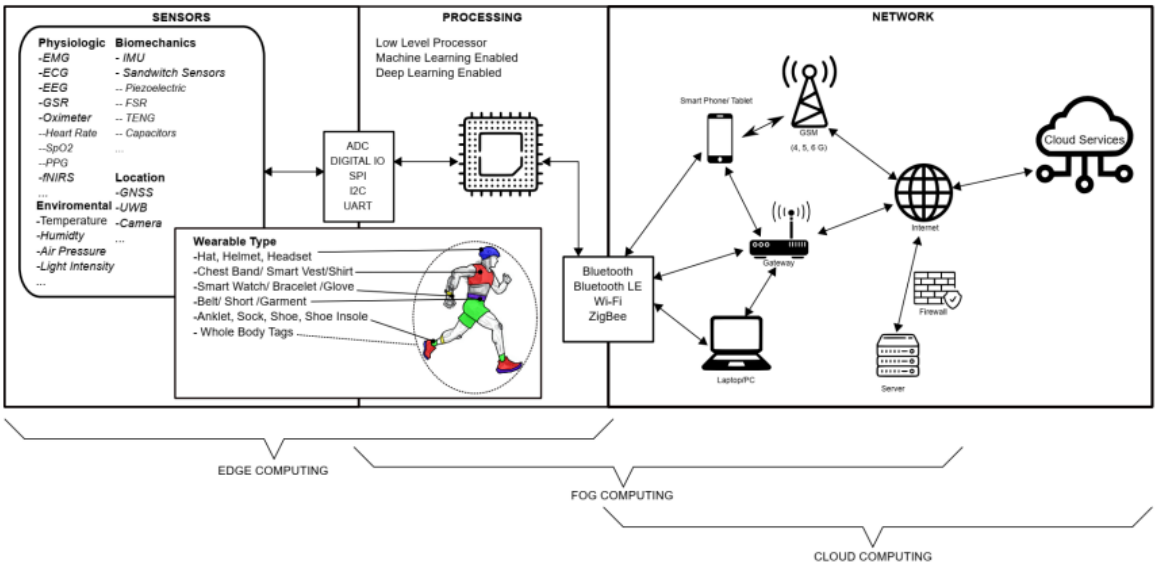


Figure 1. Layered Structure of Wearable Computing (Seçkin et al. 2023.)

Sports technology has become increasingly relevant for performance improvement. Both professional and amateur athletes rely on wearable sensors to enhance training efficiency and competition outcomes. At the same time, research highlights that wearable systems are also used for injury prevention, rehabilitation, and overall performance optimization. This means that sports electronics are not limited only to monitoring performance, but also support health-related outcomes in sporting environments. (Seçkin et al. 2023.)

During activity, continuous monitoring of physical indicators and performance parameters can enhance the safety and reliability of the whole exercise process. Such monitoring enables easier identification of abnormal conditions, supports injury prevention and allows potential risks to be detected and addressed more efficiently. Previous research has shown that injuries or fatalities caused by improper sports activity have occurred, which supports the need to improve methods of monitoring the human body during exercise. Through wearable and connected devices, real-time data monitoring can be carried out during sports, and protective measures can help in avoiding the dangers. (Zhang 2024.)

In addition to their benefits, wearable sports electronics are also associated with several challenges and limitations. Seçkin et al. (2023) highlight issues related to data accuracy, as the measurement reliability may be affected by sensor placement, motion, and environmental

conditions. On top of this the increasing collection and transmission of personal physiological and location data raises concerns regarding data privacy and security. Cost and device accessibility are also identified as limiting factors, as advanced wearable systems might not be equally available to all possible user groups. These challenges prove the need for careful system design and optimization when integrating wearable electronics into sports applications. (Seçkin et al. 2023.)

2.2 Energy Demand and Power Constraints of Modern Sports Electronics

Modern sports electronics can be considered as low-power IoT devices, whose operating lifetime is strongly constrained by limited battery capacities. Despite their smaller physical sizing, the devices are required to continuously sense, process, collect and transmit data. Without this procedure, proper health monitoring and activity tracking is impossible. The continuous use and limited battery volume in these compact devices creates the need for frequent recharging or battery changes. (Yamin & Bhat 2021.) As a result, frequent charging and battery replacement becomes a major practical limitation in real-world wearable applications (Beniwal et al. 2023.)

Afshar et al. (2025) further emphasize that the miniaturized and flexible form factors required in wearable electronics severely restrict the size of the onboard batteries. According to their review, reducing device dimensions directly limits the achievable energy density of conventional batteries. This creates a fundamental mismatch between the increasing functional complexity of wearable systems and their available energy capacity. (Afshar et al. 2025.)

In addition to storage limitations, low-power wearable devices still face substantial cumulative energy demand due to the continuous sensing, processing and wireless transmission. Even when individual components are optimized for low power consumption, the combined operation results in non-negligible energy drain over time. (Arya et al. 2024.)

Beniwal et al. also note that reliance on conventional battery-based power supply introduces reliability concerns in long-duration monitoring scenarios. Battery depletion causes interruptions in data collection, which then reduces the effectiveness of health and activity monitoring. Such interruptions are especially problematic in applications that require continuous or autonomous operation, where normal access to charging cannot be guaranteed. (Beniwal et al. 2023.)

Overall, the literature consistently identifies limited battery capacity, continuous energy demand and strict physical design limitations as key power-related challenges in modern day wearable sports electronics. These constraints show the operational boundaries of wearable devices and motivate to explore possible alternative or additional power solutions. With such innovations device lifetime could be extended without increasing battery size or compromising wearability.

2.3 Limitations of Conventional Battery-based Power Supply and Motivation for Hybrid Systems

Conventional battery-based power supplies remain the most dominant energy source in wearable electronic systems. However, extensive research has shown their fundamental limitations in supporting the long-term device operation. (Beniwal et al. 2023.) Moreover the continuing trend toward increasingly compact, lightweight and flexible wearable solutions further intensifies these limitations. As device dimensions are reduced to enhance user comfort, the available room left for energy storage decreases along the process. Although advances in battery materials and flexible battery architectures have improved energy density, the battery capacities remain constrained. On top of this, repeated charging cycles lead to gradual degradation of capacity, requiring more frequent recharging. (Yu et al. 2024.)

Interruptions in power supply caused by battery depletion present significant challenges in applications requiring continuous monitoring. Wearable sensors used for physiological data collection rely on uninterrupted operation to ensure accuracy of data. When battery energy is exhausted, monitoring processes are inevitably halted, which then compromises the continuity of collected data and therefore limits the effectiveness. Such operational disruptions highlight the mismatch between the growing functional demands of wearable electronics and the finite capabilities of conventional battery powered systems. (Beniwal et al. 2023.)

In addition to limitations in capacity and lifespan, conventional batteries pose structural and material-related challenges. The metallic materials used in batteries raise concerns regarding environmental sustainability and long-term biocompatibility in devices intended for prolonged skin contact. (Yu et al. 2024.)

These persistent limitations have motivated the development of hybrid energy systems that integrate conventional energy storage units with ambient energy harvesting technologies.

Hybrid approaches combine multiple energy sources, such as photovoltaic cells, thermoelectric generators and piezoelectric harvesters, with batteries or flexible storage devices to supplement and stabilize power supply. By converting environmental energy into usable electric power, hybrid systems reduce reliance on battery capacity while extending device autonomy and improving overall system sustainability. (Afshar et al. 2025.)

Hybrid configurations address the irregularity associated with individual energy harvesting mechanisms. Ambient energy availability varies with environmental conditions and users activity. Therefore integrating multiple harvesting methods alongside energy storage ensures more continuous and reliable energy delivery. Efficient energy management systems regulate energy conversion, storage, and distribution in order to achieve device requirements. This enhances the feasibility for long-term autonomous operation, especially in low-power wearable applications. (Kang & Yeo 2024.)

The existing literature demonstrates that constraints related to limited capacity, degradation over time, mechanical restrictions and operational reliability drive the adoption of hybrid energy systems. Incorporating ambient energy harvesting technologies provide a promising pathway toward extending device lifetimes, improved user experience, and enabling significantly long continuous operations in advanced sports and healthcare applications.

3 Solar Energy Harvesting in Small-scale Applications

Energy harvesting technologies have gained notable attention as potential power solution for wearable and low-power electronic devices due to the limitations of conventional batteries' size, sustainability and lifetime. In the case of wearable and sports electronics, harvesting energy from the surrounding environment reduces the dependence of frequent recharging alone. Previous studies identify mechanical, thermal and solar energy as the most relevant possible energy sources for wearable-scale energy harvesting. (Afshar et al. 2025; Kang & Yeo 2024.)

Among these options, solar energy harvesting is particularly attractive for outdoor wearable applications such as sports electronics. This is due to the relatively high energy density of solar radiation compared to other ambient sources. While mechanical and thermal harvesting relies on user motion and temperature, solar energy is capable of providing comparatively higher and more predictable power output under favourable environmental conditions (Kang & Yeo 2024.)

3.1 Overview of Different Energy Harvesting Methods: Mechanical, Thermal, and Solar

Wearable energy harvesting technologies can be categorized according to the type of ambient energy they convert into electrical power. Mechanical energy harvesting exploits biomechanical motion via piezoelectric and triboelectric mechanisms, that convert vibration, deformation or contact separation into electrical energy. These are mostly suited for harvesting the energy from walking, running or other repetitive body movements. Energy output is highly dependent on activity level and motion patterns, challenging the efficiency of the method. (Afshar et al. 2025; Kang & Yeo 2024.)

Thermal energy harvesting utilizes body-to-environment temperature differences. Typically happening through thermoelectric generators operating based on the Seebeck effect. While the thermoelectric harvesters can provide continuous power, their output power is generally limited to microwatt-to-milliwatt range and very strongly depends on conditions. (Kang & Yeo 2024; Teixeira et al. 2021.)

Solar energy harvesting relies on photovoltaic conversion of light into electricity and can utilize both outdoor sunlight and also indoor artificial lighting. Compared to mechanical and thermal methods, photovoltaic harvesting offers higher potential power density, especially in outdoor environments where the solar irradiance is abundant. This is the reason why photovoltaic systems are widely considered as one of the most promising energy harvesting methods for wearable electronics, especially the ones used in sports and outdoor activities. (Arya et al. 2024; Kang & Yeo 2024.)

Table 1 summarizes the typical power densities achievable from different ambient energy sources relevant to wearable applications. The comparison proves the clear advantage of ambient light under direct solar conditions, reaching up to 100 mW/cm^2, while thermoelectric

and biomechanical sources remain notably lower. This further supports the positioning of photovoltaic harvesting as the primary energy harvesting method of interest in this thesis. (Tohidinejad et al. 2024.)

Table 1. Power densities of common ambient energy sources (Tohidinejad et al. 2024.)

Energy source	Typical power density
Ambient light (direct sunlight)	100 mW/ cm^2
Ambient light (indoor illumination)	100 µW/cm^2
Thermoelectric	60 µW/cm^2
Radio frequency (ambient)	1 µW/cm^2
Radio frequency (external source)	15 µW/cm^2
Human biomechanical (microgenerator)	4 µW/cm^2
Human biomechanical (piezoelectric)	200 µW/cm^2

3.2 Factors Affecting Solar Harvesting Yield

Solar PV output is strongly shaped by environmental conditions, and understanding the influence is necessary, because the given standard rating conditions do not often represent real-world operation. Bamisile et al. (2025) note that Standard Test Conditions (STC) can overstate expected output and they contrast it with Photovoltaic USA Test Condition (PTC). PTC includes higher cell temperature and yields more conservative module ratings. (Bamisile et al. 2025.)

A key factor is solar irradiance, which has strong geographic and temporal variability. It is identified as the most important individual factor affecting the achieved PV output. Eze et al. (2024) similarly describe irradiance as the sunlight intensity reaching a panel per unit area and emphasize that changes in the irradiance directly cause fluctuations in PV power output. Major changing elements being time of day, seasons and weather. (Eze et al. 2024.)

A second central driver is the temperature. Bamisile et al. (2025) report that increasing module temperature reduces efficiency, and summarize that raising module temperature reduces the efficiency about 0.4 to 0.5 % per °C in their reviewed context. (Bamisile et al. 2025.)

Eze et al. (2024) also describe efficiency losses with rising temperature due to increased electrical resistance and lower output voltage and power. Relating this to the concept of temperature coefficient referenced to 25 °**C.**

Beyond irradiance and temperature, Bamisile et al. (2025) review additional environmental influences, including atmospheric conditions. Clouds, pollutants, dust, snow and extreme weather impacts introducing notable variability. (Bamisile et al. 2025.)

3.3 Solar Irradiance Fundamentals

Solar irradiance represents the primary variable affecting the performance of photovoltaic energy harvesting. It is commonly defined as the radiant power of sunlight intensity per unit area. Most usually expressed as watts per square meter (W/m^2). Different variations in solar irradiance directly influence the instantaneous power output of photovoltaic devices. For this reason, it is a crucial parameter for evaluating the potential of energy harvesting (Eze et al. 2024.)

Eze et al. (2024) describe solar irradiance as the intensity of sunlight reaching a surface, highlighting that any fluctuations in irradiance cause changes in power generation. Multiple temporal and environmental factors such as time of day, seasonal cycles and atmospheric conditions significantly influence the intensity and predictability of solar input. (Eze et al. 2024.)

As previously discussed, temporal variability plays a dominant role in shaping irradiance availability. Yamin & Bhat (2021) report that solar energy shows strong diurnal patterns, where mid-day irradiance levels significantly exceed the ones observed during dawn or dusk. This variation results in critically uneven harvesting over a typical day. (Yamin & Bhat 2021.)

In addition to daily cycles, seasonal variability substantially alters the availability of solar energy. Yamin & Bhat (2021) demonstrate the maximum solar irradiance differences across different seasons. Their analysis shows example peak irradiance values measured in Golden, Colorado reaching approximately 95 mW/cm^2 in July. In December the same measure decreased to about 48 mW/cm^2. This properly illustrates the strong dependence of solar harvesting potential on seasonal solar geometry and atmospheric conditions. (Yamin & Bhat 2021.)

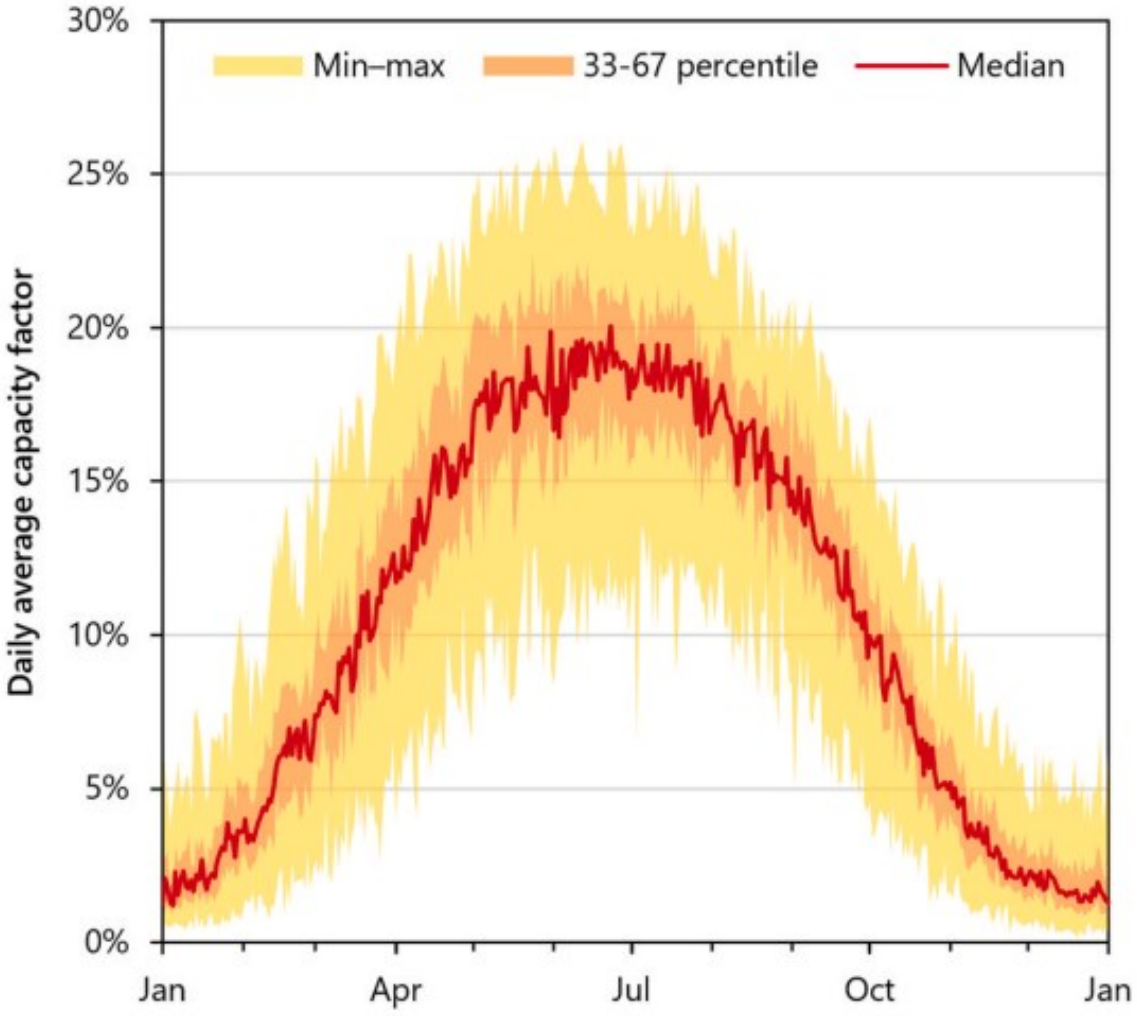


Figure 2. Temporal variability in solar PV capacity factors in the United Kingdom. Daily average capacity factor summarized over 30 years, showing year to year variability. (Bamisile et al. 2025)

The contrast between different climatic regions further illustrates how strongly geography shapes solar harvesting potential. While Yamin & Bhat (2021) report peak irradiance values in Colorado reaching 95 mW/cm^2 in July, Figure 2 represents a different picture from the United Kingdom, a high latitude, maritime climate. The capacity factor represented in the figure describes the ratio of actual average power output to the theoretical maximum under full irradiance, meaning a 20 % capacity factor indicates the system produces on average one fifth of its rated peak output over the course of a day. Based on 30 years of data, the median daily capacity factor peaks near 20 % in summer and falls close to 2 % in midwinter, with the wide min-max range reflecting the substantial day-to-day variability. From irradiance point of view,

this means the gap between instantaneous peak irradiance and realistic daily energy yield widens drastically in northern climates and winter months. Together these two datasets show how seasonal variability is not a minor correction factor, but more a defining characteristic of solar harvesting potential. This being particularly relevant for wearable devices intended for year-round use in northern latitudes. (Bamisile et al. 2025).

Atmospheric and environmental conditions introduce even more variability and uncertainty in irradiance levels. Eze et al. (2024) state, that factors such as cloud cover, environmental obstructions and floating particles are key determinants affecting the intensity of solar radiation reaching photovoltaic surfaces. These factors cause rapid and sometimes unpredictable fluctuations in harvestable solar yield. (Eze et al. 2024.)

Similarly, Yamin & Bhat (2021) note that solar energy availability is inherently random because of weather variability, geographical location and seasonally typical characteristics. The authors highlight that these fluctuations pose significant challenges particularly under conditions where the harvested energy approaches relatively low or near zero levels. (Yamin & Bhat 2021.)

Overall, solar irradiance emerges in the literature as a dominant but natural variable factor controlling photovoltaic energy harvesting. The mentioned temporal and environmental fluctuations associated with irradiance, especially in wearable context, make effective energy storage and management essential.

3.4 Typical Energy Output Ranges for Miniaturized PV Systems

The achievable energy output for miniaturized photovoltaic (PV) systems is fundamentally constrained by the limited surface area left available in small electronic applications. Unlike conventional PV installations, small wearable-integrated PV devices operate under strict geometric, mechanical and operational limitations. These inevitably restrict the possible amount of harvestable solar energy (Afshar et al. 2025).

Yamin & Bhat (2021) report that photovoltaic cells used in energy-harvesting devices can generate power densities scaling widely depending on illumination conditions. Their study indicates that PV cells can harvest approximately 0.1 mW/cm^2 under traditional indoor

lighting. With outdoor solar exposure, the power densities increase up to 100 mW/ cm^2. With such a large variation, it is easy to highlight the strong dependence on environmental irradiance levels. (Yamin & Bhat 2021).

The theoretical upper limits associated with most favourable outdoor irradiance scenarios rarely translate to reality. Arya et al. (2024) emphasize that ambient energy sources utilized in wearable electronics are naturally irregular and fluctuating.

Experimental studies on small photovoltaic modules further illustrate the limited yet usable power levels achievable in miniature systems. Shore et al. (2021) investigated several PV mini-modules under indoor LED illumination of approximately 1000 lx and reported output powers between roughly 1.36 mW and 3.25 mW depending on the photovoltaic material that was used. In their experiments, even lowest performing silicon mini module produced around 1.36 mW, while higher efficiency versions achieved 1.99 mW and 3.25 mW respectively. These results demonstrate the realistic ability to provide milliwatt-level power suitable for low-power wireless sensors and IoT devices. (Shore et al. 2021).

However Afshar et al. (2025) remind that energy harvesting technologies integrated into wearable devices are subject to wide range of application-specific constraints. The practical energy conversion efficiency and total harvested energy are strongly influenced by device miniaturization and flexibility requirements. As wearable PV surfaces are necessarily small and most often unoptimally oriented, the resulting output is limited to microwatt to milliwatt scale in most scenarios. (Afshar et al. 2025),

4 Hybrid Energy Systems for Wearable Sports Electronics

This chapter examines hybrid energy systems as a solution to the power constraints identified in the preceding chapters. The concept and definition of hybrid energy systems in wearable electronics are first introduced, followed by a proper description of typical architectures and their key components. Energy flow and storage strategies are then discussed, after which performance evaluation metrics and reported energy gains from experimental studies are presented.

4.1 Definition of Hybrid Energy Systems in Wearable Electronics

Hybrid energy systems refer to power architectures that combine multiple energy sources and storage components in order to improve the reliability and autonomy of electronic devices. In contrast to conventional battery-only systems, hybrid approaches integrate ambient energy harvesting methods together with energy storage elements such as batteries or supercapacitors. This relatively modern configuration enables devices to simultaneously harvest, store and distribute energy depending on the prevalent environmental conditions and device's individual demand. (Afshar et al. 2025.)

The main motivation for hybrid architectures originates from the inevitable limitations of single source power supply in wearable devices. Conventional batteries provide stable power but suffer from seriously limited capacity. As device functionality and sensing capabilities increase, the available battery capacity slowly becomes insufficient for continuous long-term operation (Yu et al. 2024). For this reason, combining batteries with energy harvesting technologies has become widely investigated strategy for improving device lifetime and reducing dependence on regular charging only.

In wearable electronics, hybrid energy systems typically integrate photovoltaic, thermoelectric or biomechanical energy harvesters. These harvesting mechanisms convert energy from environmental stimuli including light, temperature gradients, or body motion into electrical power. However, the availability of all the mentioned energy sources is strongly dependent on environmental conditions and user's activity and lifestyle. (Kang & Yeo 2024.)

To overcome this variability, hybrid systems incorporate various energy storage and power management components that regulate the energy flows within the device. Energy management circuits control how the harvested energy is stored and distributed to electronic components. In some cases, the energy could be buffered during periods of lower energy availability. This coordinated interaction between harvesting, storage and power regulation makes it possible for wearable devices to maintain more stable operation compared to original single energy sources systems. (Afshar et al. 2025; Beniwal et al. 2023.)

Overall hybrid energy systems are increasingly recognized as a promising modern day approach for powering wearable electronics and sports monitoring devices.

4.2 System Architecture

Hybrid energy systems in wearable applications consist of multiple functional subsystems that together enable the conversion, storage, and utilization of harvested ambient energy. A typical architecture integrates an energy harvesting unit, power management circuitry, an energy storage unit and the electronic load of the device itself. These subsystems operate together to ensure that irregular environmental energy sources can be properly converted into a stable power supply. (Afshar et al. 2025; Kang & Yeo 2024.)

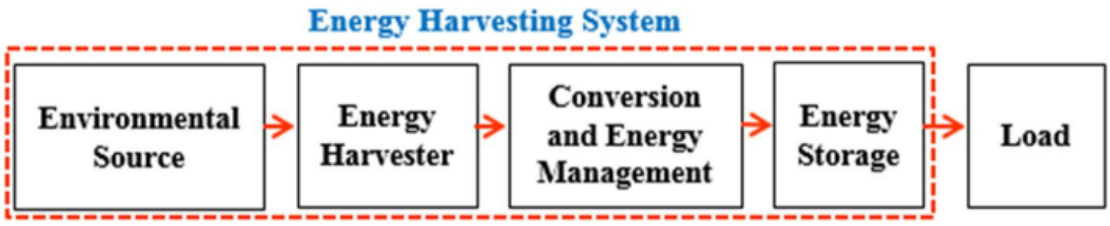


Figure 3. Simplified diagram of the architecture (Mohsen et al. 2021).

The first component being the energy harvesting unit, which converts ambient energy into electrical power. As previously argued photovoltaic harvesting is often considered particularly attractive for outdoor wearable applications such as sports watches or cycling computers. This being due to the relatively high energy density of solar radiation compared to other ambient sources and style of use of the devices. (Afshar et al. 2025; Kang & Yeo 2024.)

The electrical energy generated by the chosen harvester is typically variable and cannot be directly used to power the device in an ideally stable manner. Therefore, hybrid systems include power management circuitry that regulates the conversion and distribution of energy within the system. These circuits commonly include voltage regulation, maximum power point tracking (MPPT) and DC – DC conversion stages. These stages make the transfer of energy from harvesting unit to storage component efficient. (Yu et al. 2024.)

Next subsystem is the energy storage itself. The storage buffers periodically harvested energy and provides a stable power supply during also the insufficient periods. Batteries are the most used storage technology in these electronics due to their high energy density. Still,

supercapacitors are investigated as possible option, because of their long cycle life and fast charging ability. These storage technologies stabilize the fluctuations in harvested yield. (Afshar et al. 2025; Yu et al. 2024.)

In the end, the harvested and stored energy will be utilized by the electronic load. This includes the sensing, processing, and communication components of the device. These most often consist of various sensors, microcontrollers and wireless communication modules which make the data gathering and transmission possible. The individual components are nowadays properly optimized for low-power usage, but the combined energy demand for all of them easily exceeds the instantaneous output of ambient energy harvesters. (Arya et al. 2024.)

Power consumption of individual wearable components varies depending on their function and design. The table 2 shows representative power consumption ranges for common electronic elements used in wearables. Data processing units such as microcontrollers consume as little as 0.5 microwatts, while wireless transmission via Bluetooth or WiFi can reach up to 200 milliwatts depending on protocol. Display components represent the highest demand, with conventional OLED and LCD screens requiring hundreds to thousands of milliwatts. In some cases this alone can exceed what any ambient energy harvester can provide. Low-power display alternatives reduce this significantly, falling straight into the range of 10^{-3} to 10^{-1} mW. (Kang & Yeo 2024.)

Table 2. Power consumption of common wearable electronic elements Kang & Yeo (2024).

Component category	Example component	Typical power consumption
Data processing	MCU/SoC	0.5 µW-216 µW
Low power sensor	ECG	5.2 mW
Low power sensor	PPG	68 µW-4.8 mW
Low power sensor	Temperature sensor	82 µW-241 µW
Wireless communication	Bluetooth/WiFi/NFC	25 µW-200 mW
Display	OLED/LCD	100 mW-1000 mW
Display	Low-power display	0.1-10 mW

4.3 Energy Flow and Storage Strategies

Hybrid wearable energy systems rely on controlled energy flow between harvesting modules, storage elements and electronic loads. Because the ambient energy sources are inherently variable, hybrid systems require dedicated strategies for regulating energy transfer. This way stable device operation can be achieved. (Mohsen et al. 2021.)

A central component enabling this functionality is the power management subsystem, which governs how the harvested energy is distributed among the system. Energy generated by the harvesting units is typically conditioned before it is delivered either to energy storage components or directly to the load. This often means using rectification stages and voltage regulation circuits to ensure compatibility between electrical characteristics of harvesters and the requirements of components. (Mohsen et al. 2021.)

Different storage technologies contribute in varying ways. Rechargeable batteries typically provide higher energy density and are that way also more suitable for maintaining longer energy reserves. In contrast, supercapacitors make higher power densities possible and enables rapid charge-discharge cycles. This then enables more effective responding to rapid fluctuations in harvested energy and load demand. As a result, hybrid architectures often integrate both technologies to balance long term storage with short-term stabilization. (Afshar et al. 2025.)

More advanced energy management strategies further improve system efficiency by adapting system behaviour to changing environmental and operational conditions. Control circuits can adjust energy routing and limit the power consumption if needed. (Yu et al. 2024.)

4.4 Metrics for Performance Evaluation

The performance of energy harvesting systems is usually evaluated using several different key metrics covering energy generation, efficiency of storage and device autonomy. These work as a toolset for assessing how effectively ambient energy harvesting contributes to extending operation duration (Mohsen et al. 2021.)

Most fundamental metric is harvested power, which describes the electrical power generated by the harvesting system under given conditions. Most usually expressed in microwatts or milliwatts for wearable size applications. (Tohidinejad et al. 2024.)

Another widely used metric is energy balance. This compares the harvested energy with the energy consumed by the device. Evaluating if the harvested energy can possibly match or even exceed device consumption over time. This means that the device could theoretically operate solely without external charging. (Mohsen et al. 2021.)

Extension of device autonomy is a valid target of evaluation as well. Autonomy extension is the achieved increase in device operating time achieved through the integration of harvesting. When harvested energy does not fully sustain the device, the amount of extended battery lifetime is the measurement that matters. (Tohidinejad et al. 2024.)

Some additional performance indicators are energy conversion efficiency, storage efficiency and system reliability in more extreme environmental conditions. With these metrics, researchers and designers enhance the practical feasibility of these systems in real world applications (Mohsen et al. 2021.)

4.5 Reported Performance and Energy Gains

Experimental studies have demonstrated that photovoltaic harvesting can significantly contribute to the energy budget of wearable sensor systems. For example Tohidinejad et al. (2024) evaluated a photovoltaic-powered sensor node operating at 3.3 V with an average consumption of 32.21 mA. Under direct sunlight conditions, the photovoltaic panel produced up to 235.5 mW of power, whilst output decreased to around 140 mW when the user faced away from the sun. In shaded conditions the amount decreased all the way down to 10.25 mW. These measurements illustrate the strong dependence of conditions and user orientation.

Table 3 summarizes the measured photovoltaic output reported by Tohidinejad et al. (2024) under different orientation and shading.

Table 3. PV panel test results (Sunny day) Tohidinejad et al. (2024).

Test condition	Harvested PV Power (mW)
Direct sunlight (facing sun)	235.5
Direct sunlight (back to sun)	140
Shadow conditions	10.25

The results show that despite substantial variation in harvested power across conditions, the average device consumption of 32.21 mA remained essentially constant throughout all test scenarios. This confirms that the harvesting system must be capable of buffering energy during periods of reduced solar input, and that the capacity of storage is a critical parameter of design in any solar-assisted wearable application.

5 Strengths and Limitations

The previous chapters have covered the theoretical foundation and system level understanding of solar energy harvesting and hybrid architectures in wearable electronics. Based on this analysis, it is possible now to evaluate the practical feasibility and inherent limitations of integrating photovoltaic harvesting into real-world sports devices. While the hybrid systems seem promising for extended device autonomy, their practical performance is strongly shaped by environmental conditions, physical constraints and specific demands of use.

5.1 Practical Feasibility in Real-World Sports Environments

The translation of photovoltaic energy harvesting from controlled laboratory conditions to real-world sports use introduces a series of inevitable challenges worth noting. In experimental settings, solar cell performance is evaluated under standard test conditions. As Bamisile et al. (2025) note, these conditions can potentially overstate expected output, since real world

deployments involve variable solar angles, and atmospheric influences that are absent under lab test protocols.

In wearable sport electronics specifically, deviations from ideal conditions are the norm rather than exception. The orientation of wearable device relative to the sun changes continuously during physical activity. The measurement by Tohidinejad et al. (2024) proves this clearly. They show how photovoltaic assisted sensor node produced up to 235.5 mW when oriented toward the sun, but output decreased to approximately 140 mW when the user faced away. In the end declining to 10 mW in shaded conditions. This three value variation across realistic use scenarios demonstrates that orientation and shading are among the most significant practical constraints for wearable solar harvesting.

Beyond orientation, the type of sports activity influences the harvesting potential. Activities that include relatively stable body positions and significant solar exposure, such as road cycling or open water rowing, offer more favorable conditions for consistent photovoltaic output. On the other hand, activities involving rapid and unpredictable body movement, such as trail running or various team sports inevitably offer less ideal circumstances for harvesting solar yield.

The geographical location and season of use adds to the variability. Yamin & Bhat (2021) demonstrate that peak irradiance values measured in Colorado reached 95 mW/cm^2 in July, dropping to 48 mW/cm^2 in december. For users in higher latitude regions, the seasonal reduction introduces even more significant constraints on the annual energy yield availability.

It has also been found to be relevant to consider the current cloud cover, aerosols, dust and other atmospheric phenomena. Bamisile et al. (2025) report these being capable of reducing the electricity output by up to 60 %. While such extreme reductions are unlikely to continue throughout an entire training session, even moderate cloud cover reduces the contribution of solar harvesting to devices energy budget. These factors demonstrate that solar-assisted hybrid systems are best understood as a supplementary source under real-world sports conditions rather than a totally autonomous power solution.

5.2 Evaluation of Solar Harvesting Potential

In order to measure the realistic contribution of photovoltaic harvesting to the energy budget of a wearable sports device requires looking into the combined effect of available surface area, irradiance conditions and photovoltaic conversion efficiency. The surface area available for solar integration in wrist-worn devices is typically around few square centimetres, which constrains the harvestable power regardless of irradiance levels. Afshar et al. (2025) emphasize that wearable PV surfaces are small and most often non-optimally oriented. This resulting in practical output being limited to the microwatt to milliwatt scale in most scenarios.

Yamin & Bhat (2021) report that photovoltaic cells used in energy harvesting application can produce approximately 0.1 mW/cm^2 under typical indoor lighting, whereas outdoor solar exposure power densities can reach up to 100 mW/cm^2under absolute peak conditions. For a wearable device with a photovoltaic surface area of for example 4 cm^2 and a realistic outdoor irradiance of 50 mW/cm^2, a PV module with 15 % conversion efficiency would generate roughly 30 mW. This amount is comparable to the instantaneous power consumption of a low-power GPS enabled sports watch operating in continuous tracking mode. This suggests that under favourable outdoor conditions, solar harvesting can meaningfully offset device consumption.

However, this estimate represents near-optimal scenario. Under overcast conditions, with irradiance reduced by 50 % or more, the same device would produce less than 15 mW, insufficient to cover full device operation. The measurements reported by Shore et al. (2021) from small photovoltaic modules under indoor LED illumination of 1000 lux resulted power outputs between 1.36 mW and 3.25 mW depending on the cell material. This illustrates perfectly the severely constrained output in low-light environments. For sports devices used mostly in outdoors under direct sunlight, these figures confirm that solar harvesting can provide meaningful battery lifetime extension during active use, but cannot substitute entirely for stored battery capacity.

Temperature effects introduce further losses. Bamisile et al. (2025) report that raising module temperature reduces the efficiency by approximately 0.4 to 0.5 % per °C. For a wearable device exposed to direct sunlight, module temperatures can exceed 25 °C even at moderately common ambient conditions. This causing efficiency losses that partially offset the gains from high

irradiance. Eze et al. (2024) similarly describe how increased electrical resistance and reduced output voltage appear at elevated temperatures affecting the efficiency. In applications where the device is worn close to the skin and exposed to direct solar irradiation, thermal management of the photovoltaic surface adds complexity to system design that is possibly overlooked.

5.3 Future Development Directions

The limitations identified in previous sections point towards several directions where continued technological development is expected to improve the viability of solar assisted hybrid systems. Among the most promising areas is the advancement of flexible and textile integrated photovoltaic materials. Yu et al. (2024) note that the development of fibrous solar cells that are lightweight and flexible enables the integration with various body parts. As flexible perovskite and organic PV cells reach proper commercial maturity, the available area for solar harvesting on wearable devices could expand beyond the inflexible options that currently dominate.

Improvements in photovoltaic cell efficiency under non-ideal conditions represents another critical development area. Yu et al. (2024) highlight how a flexible quasi-two-dimensional perovskite cell module achieved conversion efficiency exceeding 31 % indoors. This enabled continuous physiological monitoring for over twelve hours without external charging. This kind of performance under very variable illumination would certainly change the energy balance calculus for wearable solar integration.

At system level, more intelligent energy management strategies are expected to improve the contribution of harvested energy. Yu et al. (2024) identify that smart power management strategies that dynamically adjust system behaviour, will play an important role in the future design of wearable devices. For sports electronics specifically, adaptive systems could possibly reduce the power consumption of non-critical functions during periods of low solar availability. During peak irradiance on the other hand, it would fully utilize harvested energy, improving autonomy and user experience.

The possible integration of multiple harvesting models alongside solar remains a relevant topic. Because solar harvesting is intermittent and weather dependent, combining it with kinetic or thermoelectric harvesting mechanisms would provide complementary energy inputs that partially compensate periods of low irradiance. Yu et al. (2024) cover that multi-source hybrid

architectures are technically achievable and can provide more stable power delivery than any single harvesting method by itself.

5.4 Implications for Sports Electronics Design

The first possibly most fundamental implication is that device placement and form factor should be selected with solar exposure in mind. As Bamisile et al. (2025) establish, irradiance is the most significant factor affecting photovoltaic output. Any design choice that reduces the effective irradiance on the harvesting surface, whether with suboptimal placement, shading created by clothing or unfavourable orientation during most typical activity, will directly reduce the contribution of solar harvesting to device autonomy.

For wrist-worn devices such as sports watches, the upper surface of the device receives consistent solar exposure during outdoor activities, making it the most practical location for photovoltaic integration. However, as the experimental measurements of Tohidinejad et al. (2024) demonstrate, output can drop drastically depending on arm rotation and cloud conditions, which adds to the need for proper energy management systems. Energy management circuits that include maximum power point tracking and dynamic load regulation would be ideal additions to solar-assisted systems to further making the operation even more practical. (Tohidinejad et al. 2024.)

Second implication concerns the appropriate framing of solar harvesting as a design element. Based on the power levels achievable under realistic worldwide conditions, solar harvesting in sports electronics is best positioned as a battery lifetime extender rather than a replacement for stored energy. Under favourable outdoor conditions, solar input can meaningfully reduce the frequency of charging cycles and support longer autonomous operation during multi-day events such as ultramarathons or expedition cycling. This framing also sets appropriate expectations, which is extremely relevant for product communication and design validation. Eze et al. (2024) note that optimal solar panel performance requires a thorough understanding of local irradiance patterns and environmental conditions. This principle being equally relevant to the design and testing of solar integration.

Finally, the environmental durability and biocompatibility requirements of sports wearables create additional constraints on photovoltaic material selection. Devices intended for prolonged skin contact during intense physical activity must withstand sweat, moisture, mechanical stress,

and UV exposure without any degradation. Yu et al. (2024) identify biocompatibility, weight, stability and toxicity as critical factors in the design of power supply modules for wearable sensing systems. These requirements are particularly important in sports applications, where the device is inevitably exposed to more demanding mechanical and environmental conditions than in typical consumer electronics use.

6. Scenario-Based Case Study

The theoretical framework established in the previous chapters provides the foundation for evaluating solar energy harvesting in a practical context. While the literature consistently identifies solar harvesting as a promising supplementary power source for wearable electronics, the gap between laboratory conditions and real-world performance remains a central challenge. This chapter applies the established concepts to three commercially available Garmin solar-assisted devices, examining how solar harvesting actually performs across realistic use scenarios. Two of the selected devices are wrist-worn wearables in the conventional sense, while the third – The Garmin Edge 1040 Solar is a handlebar mounted cycling computer and thus falls out of the strict definition of a wearable. It is nonetheless included as a comparative reference, given its near identical photovoltaic architecture, shared outdoor sports use context, and the structural advantage its fixed orientation provided relative to wrist-worn devices.

Garmin represents the most mature and widely adopted implementation of solar technology in consumer sports electronics, having integrated photovoltaic harvesting across its watch and cycling computer product lines since 2019 (DC Rainmaker 2025). The company's solar assisted devices therefore offer a uniquely well-documented case for analysis, with both manufacturer specifications and extensive real world testing data from users, available through respected product testing forums such as DC Rainmaker, one of the most reputable independent reviewers of sports technology.

The analysis includes four stages. First, the selected devices and their parameters are presented. Second, solar harvesting potential is evaluated across a range of irradiance scenarios relevant for sports use. Third, an energy balance assessment examines the relationship between harvested power and device consumption. At last, a sensitivity analysis explores how the key variables, such as the available surface area, irradiance level, usage intensity, and photovoltaic

efficiency influences the overall contribution of solar harvesting to device autonomy. Together, these steps allow for a grounded evaluation of whether solar harvesting in commercial sports electronics functions as a meaningful battery extender, or whether its practical contribution remains marginal under realistic operating conditions.

The device specifications referenced throughout this chapter are from Garmin's official product pages (Garmin 2022; Garmin 2025). Real-world performance data is sourced from DC Rainmaker's independent in-depth reviews (DC Rainmaker 2022a; 2022b; 2025), which represent among the most comprehensive publicly available testing of these devices. Origin of specific measurements are cited individually at the point of reference throughout the analysis.

6.1 Selected Devices and Parameters

Three Garmin solar-assisted devices are selected for this scenario analysis, representing different product categories, display technologies and intended use contexts within the same manufacturer's solar ecosystem. The selection allows for proper comparison across varying power demands, solar surface areas and operating conditions.

The first device is the Garmin Fenix 7 X Solar, a high-end multisport watch featuring a MIP display and the largest solar collection area within the watches. Its solar architecture consists of two distinct panel types, a highly visible rim surrounding the inner edge of the display operating at hypothetical 100 % photovoltaic efficiency, and a transparent layer beneath the sapphire glass covering the entire display area at around 7 % photovoltaic efficiency (Garmin 2026a). Garmin's official battery life claims for the solar edition assume 3 hours per day at 50 000 lux conditions, yielding a smartwatch mode duration of 37 days compared to 28 days without solar (Garmin 2026a).

The second device is the Garmin Instinct 3 Solar, representing the energy-efficiency end of the wrist-worn spectrum. Unlike the Fenix 7, the instinct 3 Solar features a revised solar architecture introduced later in the Fenix 8 Solar, where the solar collection has been concentrated solely on the bezel area surrounding the screen. Despite the reduced total surface area, Garmin and independent testing by DC Rainmaker confirm significantly increased solar capturing capability compared to the previous Instinct generation. In DC Rainmaker's static

charging tests conducted in winter sun conditions, the Instinct 3 Solar gained 20 percentage points of battery in approximately 3 hours and 45 minutes. The Memory-in-Pixel (MIP) display's extremely low power consumption means that in GPS-only mode, battery drain consistently remained below 1 % per hour in real world testing, with some 75-90 minute activities producing no measurable battery change at all. Garmin markets the Instinct 3 Solar as offering unlimited battery life in smartwatch mode under sufficient solar, exposure based on the same 50 000 lux, 3 hours per day assumption. Another test done by Wiggle, 2020 shows that 6 hours static sunlight under Garmin's measured "100 % Solar efficiency" gave the Instinct an additional 9 days of battery charge, from 3 days to 12 days. This proves that the most power efficient wrist-worn models can actually achieve unlimited device autonomy if used and sun-charged strategically. (Garmin 2026a, Garmin 2026b, DC Rainmaker 2025).

The third device is the Garmin Edge 1040 Solar, a dedicated cycling computer representing a fundamentally different product. Unlike wrist-worn devices, the Edge 1040 Solar is a handlebar-mounted and therefore maintains a consistently upward-facing orientation during use. This being a significant practical advantage for solar harvesting. Its solar architecture similarly uses two panel types, two 100 % photovoltaic efficiency strips on the bottom and top of the screen and a Power glass layer across the full display at around 15 % efficiency, a notable improvement over the 7 % achieved with Power Glass in Fenix 7 watch for example. Garmin's battery life claims for the solar edition are based on 75 000 lux conditions, a bit higher baseline than the 50 000 lux used for their watch products, but still a realistic regular sunny condition (Garmin 2026c). DC Rainmaker's real-world testing recorded 31 minutes of solar gain during a 4 hour ride under mixed cloudy and sunny conditions, and 27 minutes during a 2 hour sunny ride. Conversely, a 76 minute evening pre-sunset ride produced only 2 minutes of battery gain, illustrating the strong dependence on ambient conditions (DC Rainmaker 2022b).

Table 4 compares the reported solar-assisted battery performance characteristics of selected Garmin devices based on manufacturer specifications and independent real-world testing.

Table 4. Comparison of solar-assisted performance in selected Garmin devices (adapted from Garmin 2026 a-c; DC Rainmaker 2022a, 2022b, 2025)

Parameter	**Fenix 7X Solar**	**Instinct 3 Solar**	**Edge 1040 Solar**
Device type	Multisport watch	Outdoor watch	Cycling computer
Solar panel efficiency (rim/bezel)	100 %	100 %	100 %
Solar panel efficiency (display)	7 %	None (Only bezel)	15 %
Lux baseline for spec claims	50 000 lux	50 000 lux	75 000 lux
Solar exposure assumption	3h / day	3h / day	Daytime riding
Battery life without solar	28 days (smartwatch)	40 days (smartwatch)	35h (demanding normal mode)
Battery life with solar	37 days (smartwatch)	*Unlimited* (smartwatch)	45h (demanding normal mode)
Solar gain per hour (full sun)	-	-	+20 min / h

6.2 Solar Harvesting Potential in Selected Scenarios

Estimating the realistic solar harvesting potential of each device requires combining three key parameters, the effective photovoltaic surface area, the current irradiance conditions, and the conversion efficiency of the solar architecture. As shown in chapter 3, irradiance varies substantially with season, time of day, weather, and geographic location. The following analysis considers several representative usage conditions relevant to outdoor sports use, including sunny, cloudy and low-light examples.

The effective harvesting area differs notably between devices and directly determines the achievable power output. The Fenix 7X Solar combines the 100 % efficiency rim panel with a 7 % efficiency transparent layer beneath the sapphire glass, yielding an estimated combined effective area of approximately 2-4 cm^2 (Garmin 2026a). The Instinct 3 Solar concentrates its

revised solar architecture entirely on the bezel, eliminating the display layer but achieving significantly higher overall capture. DC Rainmaker's static tests showed 20 percentage points of battery gained in under four hours of winter sun. The edge 1040 Solar benefits from both a 100 % efficiency dedicated strip and a 15 % efficiency full display layer. Alongside that the Edge 1040 Solar also exploits its most important structural advantage, a consistently upward-facing handlebar mounted location that eliminates the orientation losses inherent in wrist-worn devices. (DC Rainmaker 2025; DC Rainmaker 2022b.)

Real-world measurements illustrate the resulting difference clearly. DC Rainmaker recorded 27-31 minutes of battery gain per 2-4 hour ride on the Edge 1040 Solar under mixed sunny conditions, and 49 minutes from two hours of static sun exposure at estimated 70 000 lux. On the other hand, over 76 minute long evening pre-sunset ride gave only 2 minutes of battery life. For the Fenix 7X Solar, Garmin's specifications translate solar contribution to 22-37 % additional runtime in GPS modes under reference conditions. The Instinct 3 Solar's sub 1 % per hour GPS-mode battery drain means even modest solar input can sustain operation, with some 75-90 minute outdoor activities producing no measurable battery change at all. (DC Rainmaker 2025; DC Rainmaker 2022a; DC Rainmaker 2022b.)

6.3 Sensitivity Analysis

The energy balance results presented in the preceding section are based on specific assumptions regarding irradiance conditions, device orientation, photovoltaic efficiency and time of usage. In practice, each of these variables fluctuates. This is why understanding their individual impact on solar harvesting is essential for evaluating the results.

Irradiance level is the single most influential variable, consistent with the findings of Bamisile et al. (2025) and Eze et al. (2024) reviewed earlier. Across all three devices, a reduction from clear summer conditions to overcast, irradiance levels reduces solar's effect by a magnitude. For the Edge 1040 Solar, DC Rainmaker's (2022b) measurements demonstrate this directly, gains of half an hour per ride under sunny conditions collapse to 2 minutes in the evening. The Instinct 3 Solar is least sensitive to irradiance reduction in absolute terms, since its base consumption is so low that even weakened solar input remains proportionally relevant.

Device orientation causes additional variability specific to wrist-worn devices. The Edge's fixed upward facing position avoids this issue almost entirely, whereas smartwatch output depends heavily on arm position and activity type. As demonstrated by Tohidinejad et al. (2024), a sensor node facing away from the sun produced approximately 40 % less output than one facing towards it. For the Fenix 7X Solar and Instinct 3 Solar, activities involving stable arm position such as road cycling offer more favourable harvesting conditions than activities with rapid and unpredictable movement such as trail running or team sports.

Photovoltaic efficiency and available surface area define the upper boundary of what any solar architecture can deliver. As Yu et al. (2024) note, flexible and textile-integrated photovoltaic materials may eventually expand effective harvesting area beyond current rigid designs. For the present generation of devices, surface area still remains a hard constraint that cannot be compensated by irradiance alone. For example a device with 2 cm^2 of effective PV area operating at 15 % efficiency under 100 000 lux, produces approximately 30 mW, regardless of how favourable other condition aspects are.

Usage intensity determines the consumption side of the energy balance and therefore shapes how significant any given solar harvesting is in practice. Across all three devices, the solar contribution is proportionally greatest in low-power modes. For the Fenix 7X Solar, maximum battery GPS mode gets extended from 213 to 578 hours with solar, almost a threefold increase (Garmin 2026a). On the other hand in all-system GPS mode the extension is only 14 extra hours (Garmin 2026a). The harvested power is fixed by environmental conditions, but user behaviour determines the consumed power. Reducing display brightness, disabling unnecessary sensing, or switching lower-accuracy GPS all shift the energy balance in favor of solar contribution without any change in harvesting conditions.

6.4 Discussion

The scenario analysis confirms that solar harvesting in commercial sports electronics functions primarily as a battery lifetime extender rather than an autonomous power source, consistent with the assessment presented in Chapter 5. However the degree of extension varies greatly across devices and conditions. The results reveal inspiring elements about how effectively solar technology translates to practical autonomy depending on device design philosophy.

The Instinct 3 Solar emerges as the device where solar harvesting has the most practical impact. Its combination of extremely low base consumption and improved solar architecture creates conditions, where under favorable summer irradiance, the energy balance approaches neutrality in low power GPS modes. DC Rainmaker's (2025) testing recorded sub 1 % per hour battery drain in GPS-only mode with solar. For users engaged in multi-day outdoor activities in summer conditions, this represents a real improvement in device capability. Not only extending runtime, but effectively removing the charging constraint altogether under right circumstances. Garmin's marketing claim of unlimited battery life in smartwatch mode, while conditional on specific irradiance assumptions, is supported by independent testing under realistic summer conditions.

The Fenix 7X Solar presents a more nuanced picture. Its higher power consumption in active GPS modes means solar contributes a meaningful but modest extension. A gain of roughly 20-40 % in typical GPS usage. The value of solar in this device is most noticeable over multi-day expeditions where cumulative gains show up, or in low-power modes where the proportional contribution rises drastically. For everyday training use involving daily GPS sessions of one to three hours, the practical impact of solar to charging frequency is modest, especially at higher latitudes where winter irradiance is severely limited.

The Edge 1040 Solar stands apart in this analysis. Its consistent upward orientation and larger solar surface area give it the most reliable harvesting geometry of the three devices. DC Rainmaker's (2022b) real-world measurements validate Garmin's claims of meaningful gains during daylight riding. For the target use case of long-distance cycling and multi-day bikepacking in summer conditions, solar harvesting properly reduces the need for external charging, representing a genuine benefit. However, outside high-irradiance conditions, the gains diminish rapidly, and the devices higher power consumption means it cannot approach energy neutrality in the way of the Instinct 3 Solar.

Across all three devices, the Finnish latitude context introduces a seasonally asymmetric constraint. Summer conditions with long daylight hours and irradiance levels that can exceed Garmin's reference benchmarks, represent the favorable end of the harvesting spectrum. Winter conditions, characterized with low sun angles and short days dramatically reduces solar contribution across all devices regardless of their architecture. This seasonal asymmetry means that solar-assisted sports electronics deliver their stated value most reliably to users who are active outdoors primarily in spring, summer and autumn. This exact limitation is rarely

highlighted in product marketing but is directly relevant to consumer expectations in northern markets.

7 Conclusions

This thesis examined the role of solar energy harvesting within hybrid energy systems for wearable sports electronics. The study addressed the growing mismatch between the increasing functional demands of modern sports wearables and the fundamental limitations of conventional battery-based power supply.

The background review established that wearable sports electronics operate as low-power IoT devices subject to strict physical design constraints. Limited battery capacity, continuous energy demand from sensing, processing and wireless transmission, and the impracticality of frequent recharging in field conditions collectively motivate the search for secondary power solutions. Among available ambient energy harvesting methods, mechanical, thermoelectric and photovoltaic, solar harvesting was identified as particularly relevant for outdoor sports applications due to its comparatively high power density under favourable irradiance conditions.

The analysis of solar harvesting fundamentals demonstrated that photovoltaic output in miniaturized systems is strongly affected by irradiance variability, device orientation, available surface area and photovoltaic conversion efficiency. Standard test conditions systematically overstate real-world performance, and seasonal variability is especially important to take into consideration at northern latitudes. These characteristics position solar harvesting as an inherently irregular energy input rather than a reliable standalone power source.

Hybrid system architectures combine photovoltaic harvesters with power management circuitry and energy storage elements such as batteries or supercapacitors. The power management subsystem, including maximum power point tracking and DC-DC conversion is essential for efficiently transferring variable energy into storage. This makes sure the delivery to the electronic load stays as stable as possible. The combined energy demand of typical wearable components such as sensors, microcontrollers, wireless modules and displays

generally exceeds the instantaneous output of ambient harvesters, thereby confirming that energy storage remains an essential system component.

The scenario analysis of three commercial Garmin solar-assisted devices provided validation of the theoretical framework. The results confirmed that solar harvesting functions primarily as battery lifetime extender rather than an autonomous power source under realistic conditions. The extent of solar contribution varied substantially across devices and operating contexts. The Garmin Instinct 3 Solar, combining extremely low base consumption with improved bezel-concentrated solar architecture, showed the closest approach to energy neutrality in low-power GPS modes under summer irradiance. The Garmin Fenix 7X Solar provided notable but modest extensions in the runtime whilst in active GPS modes, with proportionally greater gains in low power usage. The Garmin Edge 1040 Solar benefited from a structurally advantageous handlebar-mounted orientation, gathering the most consistent solar harvesting amongst the three devices. While Edge 1040 Solar is not a wearable device in the strict sense of the term, it serves as a closely related companion product sharing near-identical photovoltaic technology and outdoor sports use context, which is why it is included here as a comparative reference point.

Sensitivity analysis identified irradiance level as the single most critical variable, consistent with the literature review findings. Device orientation introduced additional variability specific to wrist-worn products, while photovoltaic efficiency and surface area defined the architectural upper boundary of achievable output. Usage intensity determined the consumption side of the energy balance, with lower power operating modes greatly benefiting from solar contribution.

The practical implications for sports electronics design emphasize that device placement, form factor and power management strategy should be selected with solar exposure in mind. Solar harvesting is most appropriately framed as a charging cycle reducer and a runtime extender. Showing up to be especially valuable for multi-day outdoor activities conducted under adequate irradiance. Seasonal asymmetry at northern latitudes means that stated solar benefits are most reliably realized during spring, summer and early autumn. Future development directions include flexible and textile integrated photovoltaic materials, higher-efficiency cells under indirect illumination and intelligent adaptive energy management strategies that maximize harvested contribution across even widely variable conditions.

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